

$$\begin{cases} 4x_1 - 3x_2 + 5x_3 = 3 \\ x_1 + x_2 - 3x_3 = 5 \\ 3x_1 - 4x_2 + 8x_3 = 6 \end{cases} \quad \begin{bmatrix} 4 & -3 & 5 & 3 \\ 1 & 1 & -3 & 5 \\ 3 & -4 & 8 & 6 \end{bmatrix}$$

$$R_1 \leftrightarrow R_2 \quad \begin{bmatrix} 1 & 1 & -3 & 5 \\ 4 & -3 & 5 & 3 \\ 3 & -4 & 8 & 6 \end{bmatrix}$$

$$-4R_1 + R_2 \rightarrow R_2 \quad \begin{bmatrix} 1 & 1 & -3 & 5 \\ 0 & -7 & 17 & -17 \\ 3 & -4 & 8 & 6 \end{bmatrix}$$

$$-3R_1 + R_3 \rightarrow R_3 \quad \begin{bmatrix} 1 & 1 & -3 & 5 \\ 0 & -7 & 17 & -17 \\ 0 & -7 & 17 & -9 \end{bmatrix}$$

$$-R_2 + R_3 \rightarrow R_3 \quad \begin{bmatrix} 1 & 1 & -3 & 5 \\ 0 & -7 & 17 & -17 \\ 0 & 0 & 0 & 8 \end{bmatrix}$$

$$-\frac{1}{7}R_2 \rightarrow R_2 \quad \begin{bmatrix} 1 & 1 & -3 & 5 \\ 0 & 1 & -17/7 & 17/7 \\ 0 & 0 & 0 & 8 \end{bmatrix}$$

The last row represents the equation $0x_1 + 0x_2 + 0x_3 = 8$, which is impossible. So the system is inconsistent.

$(x_1, x_2, x_3) = \text{No Solution}$